Environmental Product Declaration

In accordance with ISO 14025 and EN 15804:2012+A2:2019 for:

PC-strand – Prestressed steel for reinforcement of concrete

from

Hjulsbro Steel AB

EPD of multiple products, based on the average results. This EPD covers 2 products:
3-wire strand
7-wire strand

Programme:	The International EPD® System, www.environdec.com
Programme operator:	EPD International AB
Rype of EPD:	EPD of multiple products from a company
EPD registration number:	S-P-02400
Publication date:	2020-12-16
Revision date:	2025-12-02
Valid until:	2030-12-01

An EPD may be updated or depublished if conditions change. To find the latest version of the EPD and to confirm its validity, see www.environdec.com

General information

Programme information

Programme:	The International EPD® System
Address:	EPD International AB Box 210 60 SE-100 31 Stockholm Sweden
Website:	www.environdec.com
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CEN standard EN 15804 serves as the Core Product Category Rules (PCR)
Product category rules (PCR): PCR 2019:14 Construction products, version 2.0.1
PCR review was conducted by: The Technical Committee of the International EPD System. A full list of members is available on www.environdec.com. The review panel may be contacted via e-mail (support@environdec.com) Review chair: Rob Rouwette (chair), Noa Meron (co-chair)
Independent third-party verification of the declaration and data, according to ISO 14025:2006, via: ☒ **Individual EPD verification without a pre-verified LCA/EPD tool** Third party verifier: *DNV Business Assurance Italy S.r.l. Via Energy Park, 14 20871 - Vimercate (MB) - Italy* Accredited by: *Accredia, Accreditation n. 0010 Validation and Verification REV. 001*
Procedure for follow-up of data during EPD validity involves third party verifier: ☒ Yes ☐ No

The EPD owner has the sole ownership, liability, and responsibility for the EPD.

EPDs within the same product category but registered in different EPD programmes may not be comparable. For two EPDs to be comparable, they must be based on the same PCR (including the same version number) or be based on fully-aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have equivalent system boundaries and descriptions of data; apply equivalent data quality requirements, methods of data collection, and allocation methods; apply identical cut-off rules and impact assessment methods (including the same version of characterisation factors); have equivalent content declarations; and be valid at the time of comparison.

Company information

Owner of the EPD:
Hjulsbro Steel AB
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Tel: +46 (0)13-32 82 00
Organization no 556259-6899
https://hjulsbrosteel.com

Contact: Joachim Törnfeldt

LCA practitioner: Ellen Wasell, Envima AB

Description of the organisation: Hjulsbro Steel is an independent developer, producer and distributor of customized steel wire products for high-quality applications. In a state-of-the-art wire processing center Hjulsbro Steel produces PC-strand.

Product-related or management system-related certifications: Certified management systems are ISO 9001 and ISO 14001.

Name and location of production site(s): Hjulsbro Steel, Linköping, Sweden

Product information

Product name: PC-strand

UN CPC code: 412 Products of iron or steel

Product identification: Uncoated 3-wire and 7-wire steel PC-strand. 3-wire in dimensions from 6.5mm to 7.5mm, 7-wire in dimensions from 9.3mm to 15.7mm.

Product description: The product studied are low alloyed steel wire rod that is drawn to smaller dimensions and twined into strands for use in prestressed concrete structures, also known as PC-strand. Hjulsbro Steel's PC-strand product range consist of 3- and 7-wire strands made from high-quality wire rod in various steel grades depending on application. The following tables list the main characteristics of products covered by this EPD.

3-wire strand according to FprEN 10138-3

Steel name FprEN 10138-3	Diameter mm	Area mm^2	Nominal mass g/m	Tensile strength MPa	Char. Value of maximum force kN	Char. Value of 0.1% proof force kN
Y1860S3	6.5	21.2	166	1860	39.4	34.7
Y1860S3	6.9	23.4	183	1860	43.5	38.3
Y1860S3	7.5	29.0	227	1860	53.9	47.4

7-wire strand according to FprEN 10138-3

Steel name FprEN 10138-3	Diameter mm	Area mm^2	Nominal mass g/m	Tensile strength MPa	Char. Value of maximum force kN	Char. Value of 0.1% proof force kN
Y1860S7	9.3	52	406	1860	86.7	85.1
Y1860S7	12.5	93	726	1860	173	152
Y1860S7	12.9	100	781	1860	186	164
Y1860S7	15.2	139	1086	1860	254	228
Y1860S7	15.7	150	1172	1860	279	246

7-wire strand according to ASTM A416

Steel grade ASTM A416	Diameter inch	Area in2	Nominal mass lb/1000 ft	Breaking strength (min.) lbf	Yield strength 1.0% (min.) lbf
270	0.5	0.153	520	45000	37170
270	0.52	0.167	570	51700	40500

Application
The PC-Strand product range is used primarily for prefabricated concrete elements, hollow core slabs, beams, TT-slabs or railway sleepers and in post tensioning constructions such as bridges or silos.

Delivery status
The dimensions of the declared product may vary according to the particular requirements of the construction project. For transportation and storage, the PC-strands are winded into coils.

Technical information
All PC-Strands are available with either plain or indented surface. The standard tensile strength is 1860 MPa, but other strengths are available. The 3-wire strand wires are of uniform thickness. In our 7-strands, the core wire diameter is at least 3 % greater than that of the outer helical wires.

The standard geometric and mechanical properties of each coil are tested in our own laboratory in accordance with the EN ISO 15630-3 standard. Hjulsbro Steel is also constantly monitored by the relevant standards authorities, ensuring that product standards are fully met.

The PC-Strands are delivered in precision-wound coils and have excellent pay-off properties. The standard coil weight for 3-wire strands is 1.4 or 2 tons, and 3, 4 and 5 tons for 7-wire strands coils. Bespoke coils can also be manufactured to customer specifications.

LCA information

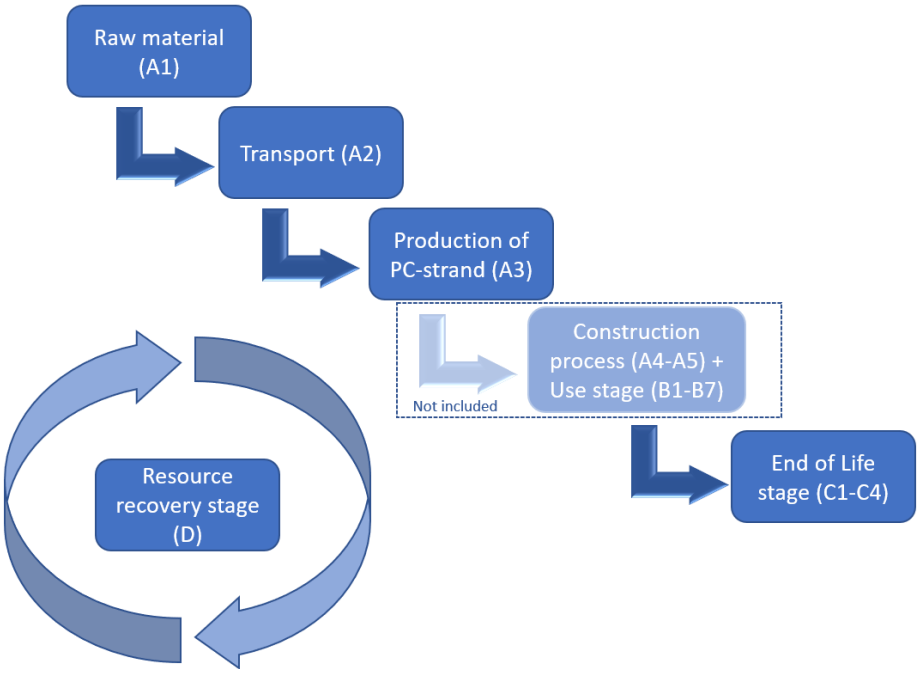
Functional unit / declared unit: 1 kg PC-strand at the factory gate plus end-of life and resource recovery stages

Time representativeness: 2024, generic data not older than 10 years.

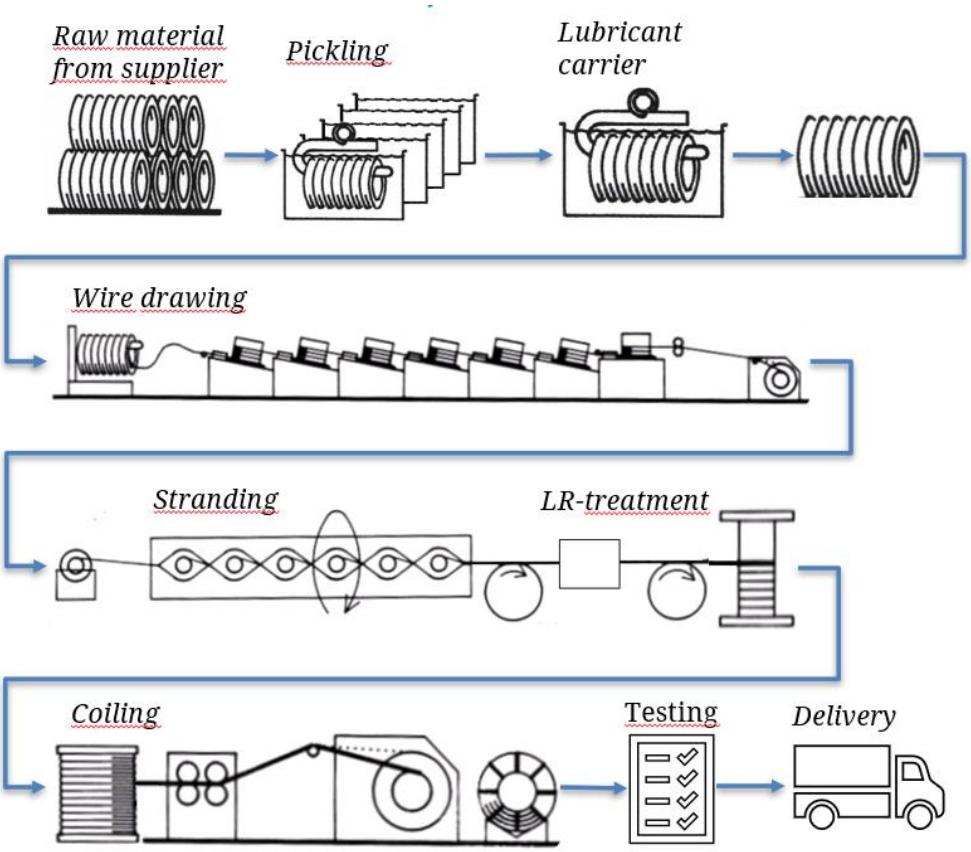
Database(s) and LCA software used: SimaPro 9.5, Ecoinvent 3.9.1

Description of system boundaries:
Cradle to gate with modules C1–C4 and module D (A1–A3 + C + D)

System diagram:



The figure below shows the entire production process for PC-strand (A3).



More information on module D:
In module D an average flow of iron scrap and pig iron was calculated to represent the recovered flow going back to the system. A material loss is considered during the recycling phase.

Summary of data-quality assessment:

Process	Source Type	Source	Reference year	Data category	Share of primary data, of GWP-GHG results for A1-A3
Manufacturing of product	Collected data	EPD Owner and Ecoinvent 3.9.1	2024	Primary data, secondary data	0,9%
Generation of electricity used in manufacturing of product	EPD, collected data, database	Ecoinvent 3.9.1	2024	Primary data	0%
Raw material production	EPD, collected data, database	EPD Owner and Ecoinvent 3.9.1	2024	Primary data, secondary data	0%
Transport in A2	Collected data, database	EPD Owner and Ecoinvent 3.9.1	2024	Primary data	5,2%
Total share of primary data, of GWP-GHG results for A1-A3					**6,1%**

The share of primary data is calculated based on GWP-GHG results. It is a simplified indicator for data quality that supports the use of more primary data, to increase the representativeness of and comparability between EPDs. Note that the indicator does not capture all relevant aspects of data quality and is not comparable across product categories.

For raw material production (A1) and electricity used in the production (A3) EPDs have been used as data. The share of primary data in these EPDs are uncertain, therefore primary data from the EPDs are set to zero.

Specific energy information
Electricity used in the core process at Hjulsbro Steel is 100% hydro power, 0.00726 kg CO2/kWh, data derives from Vattenfall EPD.

Cut-off criteria
Maintenance in the manufacturing process according to assumption that the impacts associated with these aspects are sufficiently small enough to fall below cut-off criteria of 5% of total inflows per module. Packaging related to delivery of raw materials and transport of raw material packaging also falls below the cut off criteria. Emissions released from the manufacturing (or Technosphere) to air, soil and water is considered to be very small and therefore assessed to not be affecting the results in a significant way. These data are consequently excluded in the LCA.

Data quality
Data for use in module A3 is collected directly from the manufacturing process and consist of energy consumption and recorded amount of material for the PC-strand. The data used in module A1 is

provided by suppliers of the wire rod and related components. The amount of post-consumer steel was based on the information actually available from the suppliers. The geographic region of the production site (A3) included in the calculation is Sweden. Data for A1-A2, C1-C4 and D represents the rest of the world. A2 are representing both internal and external transport. Generic data used in the EDP are not older than 10 years and site-specific data are not older than 3 years. Explanatory material may be obtained via contacting Hjulsbro Steel AB.

Methodology
This declaration is a cradle to gate with modules developed within the International EPD System and based on the application of a Life Cycle Assessment (LCA) methodology. The LCA study was performed using SimaPro 9 software.

Allocation
For certain suppliers of the raw material (wire rod), environmental impact data from existing EPDs has been applied to Module A1. For suppliers without EPDs, a conservative approach has been adopted: pre-consumer scrap is treated as virgin material, in line with the PCR requirements. When the proportion of post- and pre-consumer scrap is unknown, all scrap is assumed to be post-consumer.

To manufacture 1 kg of PC-strand, an additional input equivalent to 2.4% is required. This surplus is considered process waste and managed within the manufacturing stage.

Average product
The two different PC-strand types (3- and 7-wire) is presented as an average product, since the environmental impact differences between them are very small (6.1 % for GWP-GHG CO_2-equivalents/kg). A 3 year-average for the raw material (wire rods) has been used. This has been carried out as the variations from the suppliers can vary from year to year.

<u>Modules declared, geographical scope, share of specific data and data variation:</u>

	Product stage		Construction process stage			Use stage							End of life stage				Resource recovery stage
	Raw material supply	Transport	Manufacturing	Transport	Construction installation	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
Module	**A1**	**A2**	**A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
Modules declared	X	X	X	ND	ND	ND	ND	ND	ND	ND	ND	ND	X	X	X	X	X
Geography	SE, DE, ES US NO GB FE	SE, DE, ES US NO GB FE	SE	-	-	-	-	-	-	-	-	-	SE	SE	SE	SE	SE
Primary data	6,1%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation - products	<10%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation - sites	Not relevant			-	-	-	-	-	-	-	-	-	-	-	-	-	-

Content information

Product components	Weight, kg	Post-consumer material, weight-%	Biogenic material, weight-% and kg C/kg
Steel	1	76%	0
TOTAL	1	76%	0
Packaging materials	**Weight, kg**	**Weight-% (versus the product)**	**Weight biogenic carbon, kg C/kg**
Wood	0.00286	0.286	0.00838
Plastic (PET)	0.00009	0.009	0
Steel sheet metal band	0.00113	0.115	0
TOTAL	0.00412	0.412	0.00838

Note: 1 kg biogenic carbon is equivalent to 44/12 kg CO2.

The product contains no substances included in the REACH Candidate list (Substance of Very High Concern).

End of life scenario information

Processes	Unit	Value
Collection process	kg collected separately	1 kg
	kg collected with mixed construction waste	-
Recovery system	kg for re-use	0 kg
	kg for recycling	0.92 kg
	kg for energy recovery	0 kg
Disposal	kg product or material for final deposition	0.08 kg
Assumptions for scenario development	-	Assumption of 0.01 kWh diesel consumption per kg demolished. Assumption of 50 km distance to closest recycling facility and EURO 6 transport by truck.

Environmental Information

Note that the impact categories below are only representing the impact potentials, which means that they are approximations of environmental impacts that could occur. The average results of the included products are presented for each indicator.

Potential environmental impact – mandatory indicators according to EN 15804

Impact categories for 1 kg PC-strand. EN 15804 + A2 V1.00 / EF 3.1 normalization and weighting set.

	Results per declared unit									
	Production stage					End of life stage				Resource recovery stage
Parameter	**Unit**	**A1**	**A2**	**A3**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
GWP-total	kg CO_2 - Eq	1,21E+00	4,52E-02	1,20E-02	1,26E+00	3,57E-03	2,02E-05	1,01E-04	5,17E-06	-1,91E-03
GWP-fossil	kg CO_2 - Eq	1,20E+00	4,49E-02	1,03E-02	1,26E+00	3,57E-03	2,16E-12	1,98E-12	1,93E-13	-1,41E-09
GWP-biogenic	kg CO_2 - Eq	3,16E-03	2,59E-04	4,62E-06	3,43E-03	8,20E-07	9,31E-11	4,73E-11	4,27E-12	-9,58E-09
GWP-luluc	kg CO_2 - Eq	8,49E-04	4,64E-05	1,63E-03	2,52E-03	4,02E-07	8,81E-11	4,42E-11	3,80E-12	-9,33E-09
ODP	kg CFC11 - Eq	1,71E-08	8,72E-10	4,75E-10	1,84E-08	5,68E-11	9,25E-03	1,25E-02	8,12E-04	-4,24E-01
AP	Mol H+ Eq	4,32E-03	2,30E-04	3,24E-05	4,58E-03	3,31E-05	6,25E-02	6,30E-02	7,14E-03	-2,24E+00
EP-freshwater	kg P Eq	4,44E-05	1,15E-06	2,80E-06	4,83E-05	1,29E-08	4,50E-04	4,12E-04	2,83E-05	-6,40E-02
EP-marine	kg N Eq	9,94E-04	6,56E-05	1,10E-05	1,07E-03	1,53E-05	1,91E-03	2,88E-03	2,69E-04	-8,17E-02
EP-terrestrial	Mol N Eq	1,11E-02	7,15E-04	8,84E-05	1,19E-02	1,67E-04	6,86E-10	2,40E-09	1,29E-10	-3,66E-08
POCP	kg NMVOC - Eq	4,69E-03	2,51E-04	3,03E-05	4,98E-03	4,94E-05	9,24E-03	1,23E-02	8,02E-04	-4,23E-01
ADP-minerals & metals*	kg Sb - Eq	9,24E-06	1,46E-07	1,34E-05	2,28E-05	1,25E-09	2,05E-12	1,81E-12	2,15E-13	-2,01E-09
ADP-fossil	MJ	1,41E+01	6,70E-01	7,89E-02	1,49E+01	4,68E-02	4,21E-12	3,80E-12	4,08E-13	-3,42E-09
WDP	M^3 depriv.	1,41E-02	4,45E-03	5,64E-03	2,42E-02	1,01E-04	5,18E-05	4,52E-04	2,36E-05	-4,45E-03
Acronyms		GWP-fossil = Global Warming Potential fossil fuels; GWP-biogenic = Global Warming Potential biogenic; GWP-luluc = Global Warming Potential land use and land use change; ODP = Depletion potential of the stratospheric ozone layer; AP = Acidification potential, Accumulated Exceedance; EP-freshwater = Eutrophication potential, fraction of nutrients reaching freshwater end compartment; EP-marine = Eutrophication potential, fraction of nutrients reaching marine end compartment; EP-terrestrial = Eutrophication potential, Accumulated Exceedance; POCP = Formation potential of tropospheric ozone; ADP-minerals & metals = Abiotic depletion potential for non-fossil resources; ADP-fossil = Abiotic depletion for fossil resources potential; WDP = Water (user) deprivation potential, deprivation-weighted water consumption								

** Disclaimer: The results of the end-of-life stage (modules C1-C4) should be considered when using the results of the product stage (modules A1-A3).*

** Disclaimer: The results of this environmental impact indicator shall be used with care as the uncertainties of these results are high or as there is limited experience with the indicator.*

The estimated impact results are only relative statements, which do not indicate the endpoints of the impact categories, exceeding threshold values, safety margins and/or risks.

Biogenic carbon is actually released during stage A5, which is outside the scope of this study. Therefore, the biogenic carbon is accounted for in stage A3.

Potential environmental impact – additional mandatory and voluntary indicators

Results per declared unit										
Indicator	**Unit**	**A1**	**A2**	**A3**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
GWP-GHG*	kg CO_2 eq.	7,09E-01	4,49E-02	8,86E-03	7,63E-01	3,57E-03	9,24E-03	1,23E-02	8,02E-04	-4,23E-01
*The indicator includes all greenhouse gases included in GWP-total but excludes biogenic carbon dioxide uptake and emissions and biogenic carbon stored in the product. This indicator is thus equal to the GWP indicator originally defined in EN 15804:2012+A1:2013.										

Use of resources

Results per declared unit										
Parameter	**Unit (/kg)**	**A1**	**A2**	**A3**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
PERE	MJ	2,01E+00	7,51E-01	9,72E-02	2,86E+00	2,66E-04	2,06E-03	1,48E-02	7,39E-04	-4,06E-01
PERM	MJ	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	1,00E+00
PERT	MJ	2,01E+00	7,51E-01	9,72E-02	2,86E+00	2,66E-04	2,06E-03	1,48E-02	7,39E-04	5,94E-01
PENRE	MJ	1,42E+01	7,10E-01	7,00E-02	1,50E+01	4,97E-02	1,39E-01	2,04E-01	1,97E-02	-4,74E+00
PENRM	MJ	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	1,00E+00
PENRT	MJ	1,42E+01	7,10E-01	7,00E-02	1,50E+01	4,97E-02	1,39E-01	2,04E-01	1,97E-02	-3,74E+00
SM	kg	7,60E-01	0,00E+00	0,00E+00	7,60E-01	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
RSF	MJ	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
NRSF	MJ	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
Net use of freshwater	m^3	3,67E-03	1,76E-04	5,01E-05	3,89E-03	4,05E-06	2,20E-05	3,18E-05	2,09E-05	-4,95E-03
Acronyms	PERE = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; PERM = Use of renewable primary energy resources used as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy resources used as raw materials; PENRT = Total use of non-renewable primary energy re-sources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water									

Waste production and output flows

Results per declared unit									
Parameter	**Unit**	**A1**	**A2**	**A3**	**C1**	**C2**	**C3**	**C4**	**D**
HWD	kg	0.00E+00	0.00E+00	3.2E-02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NHWD	kg	0.00E+00	0.00E+00	1.7E-02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RWD	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
CRU	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
MFR	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
MER	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	9.20E-01	0.00E+00
EE	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Acronyms	HWD= Hazardous waste disposed; NHWD = Non hazardous waste disposed; RWD = Radioactive Waste disposed; CRU = Components for re-use; MRF = Materials for recycling; MER = Materials for energy recovery; EE= Exported energy,								

Other environmental indicators

Results per declared unit										
		Production stage					End of life - stage			Resource recovery stage
Impact category	**Unit**	**A1**	**A2**	**A3**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
Particulate Matter emissions*	disease inc.	7,95E-08	3,22E-09	1,99E-03	1,99E-03	9,23E-10	6,86E-10	2,40E-09	1,29E-10	-3,66E-08
Ionizing radiation, human health**	kBq U-235 eq	1,71E-02	1,69E-03	2,17E-04	1,90E-02	9,56E-06	6,65E-05	6,94E-04	1,75E-05	-5,26E-03
Eco-toxicity (freshwater)*	CTUe	3,44E+00	1,92E-01	1,09E-01	3,74E+00	1,68E-02	4,79E-02	4,64E-02	5,49E-03	-1,88E+00
Human toxicity, cancer effects*	CTUe	5,14E-09	3,33E-11	3,02E-11	5,20E-09	1,09E-12	4,21E-12	3,80E-12	4,08E-13	-3,42E-09
Human toxicity, non-cancer effects*	CTUe	1,09E-08	4,66E-10	1,21E-10	1,15E-08	7,61E-12	9,31E-11	4,73E-11	4,27E-12	-9,58E-09
Land use related impact/Soil quality*	Pt	3,60E+00	4,12E-01	2,26E-01	4,24E+00	3,15E-03	7,93E-02	1,92E-02	4,56E-02	-1,47E+00
Acronyms	Particulate Matter emissions= Potential incidence of disease due to PM emissions (PM) Ionizing radiation, human health= Potential Human exposure efficiency relative to U235 (IRP) Eco-toxicity (freshwater)= Potential Comparative Toxic Unit for ecosystems (ETP-fw) Human toxicity, cancer effects= Potential Comparative Toxic Unit for humans (HTP-c) Human toxicity, non-cancer effects= Potential Comparative Toxic Unit for humans (HTP-nc) Land use related impact/Soil quality= Potential soil quality index (SQI)									

** Disclaimer: The result of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experience with the indicator.*

*** Disclaimer: This impact category deals mainly with the eventual impact of low dose ionizing radiation on human health of the nuclear fuel cycle. It does not consider effects due to possible nuclear accidents, occupational exposure nor due to radioactive waste disposal in underground facilities. Potential ionizing radiation from the soil, from radon and from some construction materials is also not measured by this indicator.*

Additional LCA results (other environmental performance results) of the product(s)

The table below represents 100% recycling scenario for modules C1-C4 and D

Impact categories for 1 kg PC-strand. EN 15804 + A2 V1.00 / EF 3.1 normalization and weighting set.

Indicator	Unit	C1	C2	C3	C4	D
GWP-total*	kg CO_2 eq.	3,57E-03	9,24E-03	1,34E-02	0,00E+00	-4,60E-01
GWP-fossil*	kg CO_2 eq.	8,20E-07	8,46E-06	1,31E-04	0,00E+00	-9,82E-04
GWP-biogenic*	kg CO_2 eq.	4,02E-07	4,56E-06	9,82E-06	0,00E+00	-3,06E-04
GWP-luluc*	kg CO_2 eq.	3,57E-03	9,25E-03	1,35E-02	0,00E+00	-4,61E-01
ODP*	kg CFC 11 eq.	3,31E-05	2,02E-05	1,09E-04	0,00E+00	-2,08E-03
AP**	mol H^+ eq.	1,29E-08	7,50E-08	3,76E-07	0,00E+00	-2,95E-05
EP-freshwater*	kg P eq	1,53E-05	4,97E-06	4,50E-05	0,00E+00	-4,44E-04
EP-marine*	kg N eq.	1,67E-04	5,18E-05	4,91E-04	0,00E+00	-4,84E-03
EP-terrestrial*	mol N eq.	4,94E-05	3,13E-05	1,46E-04	0,00E+00	-2,19E-03
POCP*	kg NMVOC eq.	5,68E-11	2,01E-10	2,20E-10	0,00E+00	-8,04E-09
ADP-minerals&metals**	kg Sb eq.	1,25E-09	3,02E-08	1,04E-08	0,00E+00	-3,31E-06
ADP-fossil**	MJ	4,68E-02	1,31E-01	2,10E-01	0,00E+00	-4,87E+00
WDP**	m^3	1,01E-04	5,41E-04	1,11E-03	0,00E+00	3,36E-02
Acronyms	GWP-fossil = Global Warming Potential fossil fuels; GWP-biogenic = Global Warming Potential biogenic; GWP-luluc = Global Warming Potential land use and land use change; ODP = Depletion potential of the stratospheric ozone layer; AP = Acidification potential, Accumulated Exceedance; EP-freshwater = Eutrophication potential, fraction of nutrients reaching freshwater end compartment; EP-marine = Eutrophication potential, fraction of nutrients reaching marine end compartment; EP-terrestrial = Eutrophication potential, Accumulated Exceedance; POCP = Formation potential of tropospheric ozone; ADP-minerals&metals = Abiotic depletion potential for non-fossil resources; ADP-fossil = Abiotic depletion for fossil resources potential; WDP = Water (user) deprivation potential, deprivation-weighted water consumption					

** Disclaimer: The results of the end-of-life stage (modules C1-C4) should be considered when using the results of the product stage (modules A1-A3).*

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The table below represents 100% landfil scenario for modules C1-C4 and D

Impact categories for 1 kg PC-strand. EN 15804 + A2 V1.00 / EF 3.1 normalization and weighting set.

Indicator	Unit	C1	C2	C3	C4	D
GWP-total*	kg CO_2 eq.	3,57E-03	9,24E-03	1,40E-02	1,00E-02	0,00E+00
GWP-fossil*	kg CO_2 eq.	8,20E-07	8,46E-06	1,31E-04	1,21E-04	0,00E+00
GWP-biogenic*	kg CO_2 eq.	4,02E-07	4,56E-06	9,89E-06	2,26E-06	0,00E+00
GWP-luluc*	kg CO_2 eq.	3,57E-03	9,25E-03	1,42E-02	1,02E-02	0,00E+00
ODP*	kg CFC 11 eq.	3,31E-05	2,02E-05	1,15E-04	6,47E-05	0,00E+00
AP**	mol H^+ eq.	1,29E-08	7,50E-08	3,78E-07	7,90E-08	0,00E+00
EP-freshwater*	kg P eq	1,53E-05	4,97E-06	4,77E-05	2,73E-05	0,00E+00
EP-marine*	kg N eq.	1,67E-04	5,18E-05	5,21E-04	2,95E-04	0,00E+00
EP-terrestrial*	mol N eq.	4,94E-05	3,13E-05	1,55E-04	1,11E-04	0,00E+00
POCP*	kg NMVOC eq.	5,68E-11	2,01E-10	2,31E-10	2,94E-10	0,00E+00
ADP-minerals&metals**	kg Sb eq.	1,25E-09	3,02E-08	1,06E-08	1,90E-08	0,00E+00
ADP-fossil**	MJ	4,68E-02	1,31E-01	2,18E-01	2,32E-01	0,00E+00
WDP**	m^3	1,01E-04	5,41E-04	1,12E-03	1,02E-03	0,00E+00
Acronyms	GWP-fossil = Global Warming Potential fossil fuels; GWP-biogenic = Global Warming Potential biogenic; GWP-luluc = Global Warming Potential land use and land use change; ODP = Depletion potential of the stratospheric ozone layer; AP = Acidification potential, Accumulated Exceedance; EP-freshwater = Eutrophication potential, fraction of nutrients reaching freshwater end compartment; EP-marine = Eutrophication potential, fraction of nutrients reaching marine end compartment; EP-terrestrial = Eutrophication potential, Accumulated Exceedance; POCP = Formation potential of tropospheric ozone; ADP-minerals&metals = Abiotic depletion potential for non-fossil resources; ADP-fossil = Abiotic depletion for fossil resources potential; WDP = Water (user) deprivation potential, deprivation-weighted water consumption					

** Disclaimer: The results of the end-of-life stage (modules C1-C4) should be considered when using the results of the product stage (modules A1-A3).*

** Disclaimer: The results of this environmental impact indicator shall be used with care as the uncertainties of these results are high or as there is limited experience with the indicator.*

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